IDAI 610 – Problem Set 3

Problem 1: Inference using Full Joint Distributions

The given full joint distribution over *toothache* (t), *cavity* (c), and *catch* (k) is:

	toothache		¬toothache	
	catch	¬catch	catch	¬catch
cavity	0.108	0.012	0.072	0.008
¬cavity	0.016	0.064	0.144	0.576

Q1(a) $P(\textbf{toothache})$

The marginal probability is obtained by summing over all values of *cavity* and *catch*:

$$P(t) = \sum_{c \in \{c, \neg c\}} \sum_{k \in \{k, \neg k\}} P(t, c, k) = (0.108 + 0.012) + (0.016 + 0.064) = 0.20.$$

Hence, $P(t) = 0.20$ and $P(\neg t) = 0.80$.

Q1(b) $P(\textbf{Cavity})$

Since the variable is capitalized, we return a tuple:

$$P(\textbf{Cavity}) = \langle P(c),\, P(\neg c) \rangle,$$

where

$$P(c) = 0.108 + 0.012 + 0.072 + 0.008 = 0.20, \quad P(\neg c) = 0.80.$$

Thus, $P(\textbf{Cavity}) = \langle 0.20, 0.80 \rangle$.

Q1(c) $P(\textbf{Toothache} \mid \textbf{cavity})$

$$P(t \mid c) = \frac{P(t, c)}{P(c)} = \frac{0.108 + 0.012}{0.20} = 0.60, \quad P(\neg t \mid c) = 0.40.$$

Hence $P(\textbf{Toothache} \mid c) = \langle 0.60, 0.40 \rangle$.

Q1(d) $P(\textbf{Cavity} \mid \textbf{toothache} \vee \textbf{catch})$

Let $A = (t \vee k)$. Then

$$P(c \mid A) = \frac{P(c \wedge A)}{P(A)}, \quad P(A) = 1 - P(\neg t \wedge \neg k).$$

From the table,

$$P(\neg t \wedge \neg k) = 0.008 + 0.576 = 0.584, \quad P(A) = 0.416.$$

Also $P(c \wedge A) = 0.200 - 0.008 = 0.192$ and $P(\neg c \wedge A) = 0.800 - 0.576 = 0.224$. Hence

$$P(c \mid A) = \frac{0.192}{0.416} = \frac{6}{13} \approx 0.4615, \quad P(\neg c \mid A) = \frac{0.224}{0.416} = \frac{7}{13} \approx 0.5385.$$

Therefore $P(\textbf{Cavity} \mid t \vee k) = \langle 0.4615, 0.5385 \rangle$.

Q1(e) $P(\textbf{Catch} \mid \textbf{cavity})$

$$P(k \mid c) = \frac{0.108 + 0.072}{0.20} = 0.90, \quad P(\neg k \mid c) = 0.10.$$

Thus $P(\textbf{Catch} \mid c) = \langle 0.90, 0.10 \rangle$.

Q1(f) $P(\textbf{Catch} \mid \textbf{cavity} \wedge \textbf{toothache})$

$$P(k \mid c, t) = \frac{0.108}{0.108 + 0.012} = 0.90, \quad P(\neg k \mid c, t) = 0.10.$$

Hence $P(\textbf{Catch} \mid c, t) = \langle 0.90, 0.10 \rangle$.

Comparison. The results in (e) and (f) are identical, showing that *toothache* and *catch* are conditionally independent given *cavity*:

$$P(k \mid c, t) = P(k \mid c) = 0.90.$$

Problem 2: Naive Bayes Classification Modeling

We model **Shortlisted** $\in \{\text{True (T)}, \text{False (F)}\}$ using the Naive Bayes assumption over features:

$$P(y \mid x) \;\propto\; P(y) \prod_i P(x_i \mid y), \qquad P(y \mid x) \;=\; \frac{P(y) \prod_i P(x_i \mid y)}{\sum_{y' \in \{T,F\}} P(y') \prod_i P(x_i \mid y')}.$$

Counts and probabilities below are computed from Table 2 (14 players).

Class priors

$$P(T) = \frac{8}{14} = 0.5714286, \qquad P(F) = \frac{6}{14} = 0.4285714.$$

Class-conditional likelihoods (from Table 2)

Feature	Value	$P(\cdot \mid T)$	$P(\cdot \mid F)$
League	LaLiga	$4/8 = 0.50$	$1/6 \approx 0.1667$
	SerieA	$3/8 = 0.375$	$2/6 \approx 0.3333$
	PremierLeague	$1/8 = 0.125$	$3/6 = 0.50$
Position	CF	$5/8 = 0.625$	$1/6 \approx 0.1667$
	RW	$2/8 = 0.25$	$2/6 \approx 0.3333$
	LW	$1/8 = 0.125$	$3/6 = 0.50$
Preferred foot	Right	$6/8 = 0.75$	$1/6 \approx 0.1667$
	Left	$2/8 = 0.25$	$5/6 \approx 0.8333$
Capped	yes	$4/8 = 0.50$	$3/6 = 0.50$
	no	$4/8 = 0.50$	$3/6 = 0.50$

Q2(i): (League = LaLiga, Position = RW, Preferred = Right, Capped = yes)

Unnormalized posteriors:

$$\begin{aligned} P(T \mid x) &\propto P(T)\, P(\text{LaLiga} \mid T)\, P(\text{RW} \mid T)\, P(\text{Right} \mid T)\, P(\text{yes} \mid T) \\ &= \left(\tfrac{8}{14}\right)\left(\tfrac{4}{8}\right)\left(\tfrac{2}{8}\right)\left(\tfrac{6}{8}\right)\left(\tfrac{4}{8}\right) = \frac{3}{112} \approx 0.0267857, \\ P(F \mid x) &\propto P(F)\, P(\text{LaLiga} \mid F)\, P(\text{RW} \mid F)\, P(\text{Right} \mid F)\, P(\text{yes} \mid F) \\ &= \left(\tfrac{6}{14}\right)\left(\tfrac{1}{6}\right)\left(\tfrac{2}{6}\right)\left(\tfrac{1}{6}\right)\left(\tfrac{3}{6}\right) = \frac{1}{504} \approx 0.0019841. \end{aligned}$$

Normalization:

$$P(T \mid x) = \frac{0.0267857}{0.0267857 + 0.0019841} \approx \boxed{0.9310}, \qquad P(F \mid x) \approx \boxed{0.0690}.$$

Decision & interpretation: Shortlisted = True. LeagueLaLiga, RW, Right, and Cappedyes align strongly with the True class in this dataset, yielding a high posterior for T. **Trust:** Reasonably high for this specific combination (good support under T and weak under F), but still limited by the very small sample size (n=14) and Naive conditional-independence assumptions.

Q2(ii): (League = PremierLeague, Position = LW, Preferred = Left, Capped = no)

Unnormalized posteriors:

$$\begin{aligned} P(T \mid x) &\propto \left(\tfrac{8}{14}\right)\left(\tfrac{1}{8}\right)\left(\tfrac{1}{8}\right)\left(\tfrac{2}{8}\right)\left(\tfrac{4}{8}\right) = \frac{1}{896} \approx 0.0011161, \\ P(F \mid x) &\propto \left(\tfrac{6}{14}\right)\left(\tfrac{3}{6}\right)\left(\tfrac{3}{6}\right)\left(\tfrac{5}{6}\right)\left(\tfrac{3}{6}\right) = \frac{5}{112} \approx 0.0446429. \end{aligned}$$

Normalization:

$$P(T \mid x) = \frac{0.0011161}{0.0011161 + 0.0446429} \approx \boxed{0.02439}, \qquad P(F \mid x) \approx \boxed{0.97561}.$$

Decision & interpretation: Shortlisted = False. PremierLeague+LW+Left are common in the False class in this dataset; hence, the posterior strongly favors F. **Trust:** High for this feature combination within this toy dataset (clear skew in likelihoods toward F), but subject to the same caveats: small n, possible sampling bias, and the Naive independence assumption.

Problem 3: Text Classification with Naive Bayes

Q3. Implementation and Preprocessing

A multinomial Naive Bayes classifier was implemented entirely from scratch in Python using only standard libraries (`csv`, `math`, `re`). Each document is tokenized using a regular expression retaining only alphabetic characters and apostrophes (so words like "don't" remain a single token) and then lower-cased. The model computes empirical priors and conditional likelihoods based on token counts per class.

$$P(c \mid x) \propto P(c) \prod_{w \in x} P(w \mid c)^{tf(w,x)}, \quad \log P(c \mid x) = \log P(c) + \sum_{w \in x} tf(w,x) \log P(w \mid c)$$

All calculations are performed in log-space to avoid numeric underflow. Unseen (out-of-vocabulary) tokens in the test set are ignored—they contribute nothing to the sum. Two optional preprocessing controls were added:

- **Document-frequency pruning** (*min_df*=2) to remove tokens appearing in only one document.
- **Stop-word removal** (basic English list) to drop extremely frequent non-informative words; applied only to the 20 Newsgroups dataset.

Laplace smoothing is toggled via a boolean flag:

$$P(w \mid c) = \frac{\text{count}(w, c) + 1}{T_c + |V|},$$

where T_c is the total token count in class c and $|V|$ the vocabulary size. This replaces zero probabilities with small non-zero values and re-normalizes the distribution.

Q4. Results and Evaluation Metrics (Unsmoothed Models)

We report **accuracy**, **macro-precision**, **macro-recall**, and the **confusion matrix** for each dataset using the baseline *unsmoothed* multinomial Naive Bayes model. Rows represent actual labels, and columns represent predicted labels.

Movie Reviews (Unsmoothed Baseline) Vocabulary size $|V| = 2502$; Accuracy = 0.3933; Macro-Precision = 0.3918; Macro-Recall = 0.3933.

	neg	**pos**
neg	68	82
pos	100	50

20 Newsgroups Subset (Unsmoothed Baseline) Vocabulary size $|V| = 26{,}632$; Accuracy = 0.1780; Macro-Precision = 0.1781; Macro-Recall = 0.1918.

	misc	**christian**	**atheism**
misc	25	20	64
christian	88	12	78
atheism	66	35	39

Analysis of Baseline Results The unsmoothed models perform poorly across both datasets, particularly for the 20 Newsgroups subset. In this implementation, out-of-vocabulary tokens encountered during testing are ignored rather than assigned zero probability, preventing complete likelihood collapse. However, the absence of smoothing still weakens the model's ability to recognize classes whose informative words were unseen during training, as those tokens contribute no positive evidence to the log-likelihood sum. This effect is especially severe for the 20 Newsgroups data due to its large and sparse vocabulary. Even for the smaller Movie Reviews dataset, the lack of smoothing results in low recall across both classes, indicating that many test samples contain novel or rare words that provide no support for either class under the unsmoothed model.

Q5. Dataset Comparison and Analysis

The two datasets differ fundamentally in linguistic structure and class separability. The Movie Reviews dataset focuses on binary sentiment (positive vs. negative), with a relatively small vocabulary dominated by opinionated adjectives and adverbs such as "good", "excellent", "boring", or "terrible." Because sentiment often hinges on these explicit polarity markers, the model relies on a small number of highly informative words. However, without smoothing, the model cannot generalize to new adjectives or unseen synonyms, leading to moderate performance (39% accuracy).

In contrast, the 20 Newsgroups dataset involves topical classification with three classes—*atheism*, *christian*, and *misc*. The vocabulary is extremely large (over 26k unique words) and highly sparse: many tokens appear in only one or two documents. The unsmoothed model therefore suffers from severe data sparsity, as unseen terms dominate the test data, yielding near-random predictions (18% accuracy). This contrast illustrates how Naive Bayes' performance depends heavily on the relationship between vocabulary density and class homogeneity. Datasets with shorter, sentiment-rich texts (Movie Reviews) are more forgiving than those requiring topical inference (20NG), which require additional preprocessing or smoothing to be effective.

Q6. Critical Discussion: Advantages and Disadvantages of Naive Bayes for Text Classification

Naive Bayes remains one of the most practical and interpretable baselines for text classification. Its computational simplicity arises from the *bag-of-words* assumption, which treats each word as conditionally independent given the class label. This allows efficient parameter estimation through simple frequency counts, making the algorithm ideal for large, sparse, and high-dimensional text datasets. In this experiment, training was nearly instantaneous even with tens of thousands of vocabulary items. Moreover, each class's likelihood can be directly traced to word frequency, providing transparent explanations for why a text is classified as positive or negative.

However, the same independence assumption that enables efficiency also constrains representational power. Real-world language exhibits strong dependencies—word order, negation, and phrase-level context—that Naive Bayes cannot capture. Expressions such as "not good" or "no evidence against" demonstrate cases where polarity or meaning depends on multi-word interaction. Additionally, the model's reliance on raw word frequencies makes it highly sensitive to data preprocessing: unseen words provide no evidence for a class, while noisy rare tokens or overly frequent stopwords can distort likelihoods. Consequently, Naive Bayes performs best when supported by well-curated vocabularies and smoothing methods, but falls short compared to more modern models like logistic regression or transformer-based classifiers that capture contextual dependencies. Nevertheless, its combination of interpretability, speed, and surprisingly competitive accuracy after minimal adjustments makes it a durable benchmark in NLP.

Q7. Laplace Smoothing and Its Effect

Following the baseline experiments, Laplace (add-one) smoothing was introduced to mitigate the impact of unseen words and stabilize probability estimation. The smoothed likelihood for each word w given class c is computed as:

$$P(w \mid c) = \frac{\text{count}(w, c) + 1}{T_c + |V|}$$

where T_c is the total number of tokens in class c and $|V|$ is the vocabulary size. Adding one to every count ensures each word—including previously unseen ones—receives a small non-zero probability, allowing the model to generalize better to unseen text.

Movie Reviews (Laplace Smoothed) Vocabulary size $|V| = 2502$; Accuracy = 0.7967; Macro-Precision = 0.7967; Macro-Recall = 0.7967.

	neg	**pos**
neg	119	31
pos	30	120

20 Newsgroups (Laplace + *min_df*=2 + Stopword Removal) Vocabulary size $|V| = 7{,}634$; Accuracy = 0.7213; Macro-Precision = 0.7142; Macro-Recall = 0.6926.

	misc	**christian**	**atheism**
misc	51	34	24
christian	11	148	19
atheism	12	19	109

Discussion Smoothing substantially improved results across both datasets. For Movie Reviews, accuracy rose from 39% to nearly 80%, as unseen adjectives and infrequent sentiment terms no longer failed to contribute to class likelihoods. The improvement is even more pronounced in 20 Newsgroups (18% $\rightarrow$ 72%), where Laplace smoothing was combined with vocabulary pruning (*min_df*=2) and stopword removal. Together, these changes mitigated data sparsity, reduced vocabulary noise, and stabilized likelihoods.

While smoothing slightly redistributes probability mass away from highly frequent words, its benefit in handling unseen vocabulary far outweighs any minor loss in precision. In practical terms, Laplace smoothing transforms Naive Bayes from a fragile count-based model into a robust probabilistic classifier that generalizes effectively to real-world, high-dimensional text data.

Q2 (Bonus): Effect of Laplace Smoothing

The below equation defines the class-conditional likelihood for a word w under class c in the standard multinomial Naive Bayes model as:

$$P(w \mid c) = \frac{\text{count}(w, c)}{T_c},$$

where $\text{count}(w, c)$ is the frequency of word w in all training documents of class c, and T_c is the total number of word tokens in that class. This formulation assumes the training corpus perfectly represents the entire language distribution of each class. In reality, this assumption is unrealistic: many valid words may never appear in the limited training data, causing their conditional probability to become zero and making $P(c \mid x)$ vanish for any test document containing such words.

Change Introduced by Laplace Smoothing Laplace (add-one) smoothing modifies above equation to:

$$P(w \mid c) = \frac{\text{count}(w, c) + 1}{T_c + |V|}.$$

Here, a constant value of 1 is added to every observed count. This change ensures that even unseen words receive a small non-zero probability, while maintaining the normalization of the probability distribution. The denominator is increased by $|V|$ to compensate for the $+1$ added to each of the $|V|$ vocabulary items, guaranteeing that $\sum_{w \in V} P(w \mid c) = 1$.

Interpretation and Significance Adding 1 to each count can be interpreted as placing a *uniform prior* over all words before any data are observed. From a Bayesian perspective, this reflects prior belief that every word has been seen at least once, making the model less sensitive to sampling gaps in the training data. Practically, it prevents zero probabilities, allowing the classifier to handle new vocabulary gracefully and making the model's predictions more realistic for unseen text.

Why the denominator includes $|V|$. Laplace smoothing adds one pseudo-count to *every* word $w \in V$: $\text{count}(w, c) \mapsto \text{count}(w, c) + 1$. Since there are $|V|$ words, the total mass for class c increases by $|V|$, i.e., $T_c \mapsto T_c + |V|$. The smoothed likelihood

$$P(w \mid c) = \frac{\text{count}(w, c) + 1}{T_c + |V|}$$

therefore normalizes correctly:

$$\sum_{w \in V} P(w \mid c) = \frac{\sum_w (\text{count}(w, c) + 1)}{T_c + |V|} = \frac{T_c + |V|}{T_c + |V|} = 1.$$

Q8. Full Joint Probability Distribution

The Bayes network in Figure 1 (in problem set) consists of four Boolean variables: *Cloudy (C)*, *Sprinkler (S)*, *Rain (R)*, and *WetGrass (W)*. The structure implies the following factorization:

$$P(C, S, R, W) = P(C)\, P(S \mid C)\, P(R \mid C)\, P(W \mid S, R).$$

The conditional probability tables are:

$$\begin{aligned}
P(C{=}T) &= 0.5, \quad P(C{=}F) = 0.5, \\
P(S{=}T \mid C{=}T) &= 0.1, \quad P(S{=}T \mid C{=}F) = 0.5, \\
P(R{=}T \mid C{=}T) &= 0.80, \quad P(R{=}T \mid C{=}F) = 0.20, \\
P(W{=}T \mid S, R) &= \begin{cases} 0.99 & \text{if } S{=}T, R{=}T \\ 0.90 & \text{if } S{=}T, R{=}F \\ 0.90 & \text{if } S{=}F, R{=}T \\ 0.01 & \text{if } S{=}F, R{=}F \end{cases}
\end{aligned}$$

Substituting these values, we compute the joint probability for every combination of (C, S, R, W) using:

$$P(C, S, R, W) = P(C)\, P(S \mid C)\, P(R \mid C)\, P(W \mid S, R).$$

C	S	R	W	$P(C,S,R,W)$
T	T	T	T	0.0396
T	T	T	F	0.0004
T	T	F	T	0.0090
T	T	F	F	0.0010
T	F	T	T	0.3240
T	F	T	F	0.0360
T	F	F	T	0.0009
T	F	F	F	0.0891
F	T	T	T	0.0495
F	T	T	F	0.0005
F	T	F	T	0.1800
F	T	F	F	0.0200
F	F	T	T	0.0450
F	F	T	F	0.0050
F	F	F	T	0.0020
F	F	F	F	0.1980

A sanity check confirms that the probabilities sum to 1.0000, verifying that the joint distribution is properly normalized.

Q9. Is rain or sprinkler more likely given that the grass is wet?

We compare $P(R{=}T \mid W{=}T)$ and $P(S{=}T \mid W{=}T)$ using

$$P(A \mid W) = \frac{P(A,W)}{P(W)}.$$

Computing $P(W{=}T)$. From the full joint table in Q8, summing all entries with $W{=}T$ yields

$$P(W{=}T) = 0.6500.$$

Computing $P(R{=}T, W{=}T)$. Sum the four rows where $R{=}T$ and $W{=}T$:

$$\begin{aligned} P(R{=}T, W{=}T) &= P(C{=}T, S{=}T, R{=}T, W{=}T) \\ &\quad + P(C{=}T, S{=}F, R{=}T, W{=}T) \\ &\quad + P(C{=}F, S{=}T, R{=}T, W{=}T) \\ &\quad + P(C{=}F, S{=}F, R{=}T, W{=}T) \\ &= 0.0396 + 0.3240 + 0.0495 + 0.0450 \\ &= 0.4581. \end{aligned}$$

Thus,

$$P(R{=}T \mid W{=}T) = \frac{0.4581}{0.6500} = 0.7048.$$

Computing $P(S{=}T, W{=}T)$. Sum the four rows where $S{=}T$ and $W{=}T$:

$$\begin{aligned} P(S{=}T, W{=}T) &= P(C{=}T, S{=}T, R{=}T, W{=}T) \\ &\quad + P(C{=}T, S{=}T, R{=}F, W{=}T) \\ &\quad + P(C{=}F, S{=}T, R{=}T, W{=}T) \\ &\quad + P(C{=}F, S{=}T, R{=}F, W{=}T) \\ &= 0.0396 + 0.0090 + 0.0495 + 0.1800 \\ &= 0.2781. \end{aligned}$$

Therefore,

$$P(S{=}T \mid W{=}T) = \frac{0.2781}{0.6500} = 0.4278.$$

Conclusion. Given that the grass is wet, rain is more likely than the sprinkler:

$$P(R{=}T \mid W{=}T) \approx 0.7048 \;>\; P(S{=}T \mid W{=}T) \approx 0.4278.$$

Q10. Building the Bayes Network in Hugin-Lite and Inference Results

Network construction. A Bayesian network was created in Hugin-Lite using four Boolean nodes: *Cloudy (C)*, *Sprinkler (S)*, *Rain (R)*, and *WetGrass (W)*. The directed edges were drawn according to the model structure: $C \to S$, $C \to R$, and both S and R pointing to W. Each node was assigned its conditional probability table (CPT) exactly as given in the assignment.

Prior probabilities (no evidence). After compiling the network, the prior marginals obtained in Hugin were:

$$P(S{=}T) = 0.30, \quad P(R{=}T) = 0.50, \quad P(W{=}T) = 0.65.$$

These values are consistent with the manual computation in Q8 using the factorization $P(C, S, R, W) = P(C)P(S \mid C)P(R \mid C)P(W \mid S, R)$. Figure 1 shows the implemented Bayesian network with the prior marginals.

Posterior with evidence $W{=}T$. When evidence was entered for *WetGrass* = `True`, the posterior probabilities updated to:

$$P(R{=}T \mid W{=}T) \approx 0.7048, \qquad P(S{=}T \mid W{=}T) \approx 0.4278.$$

These match the manually computed results from Q9, confirming the correctness of the model inference. The updated posterior view is shown in Figure 2.

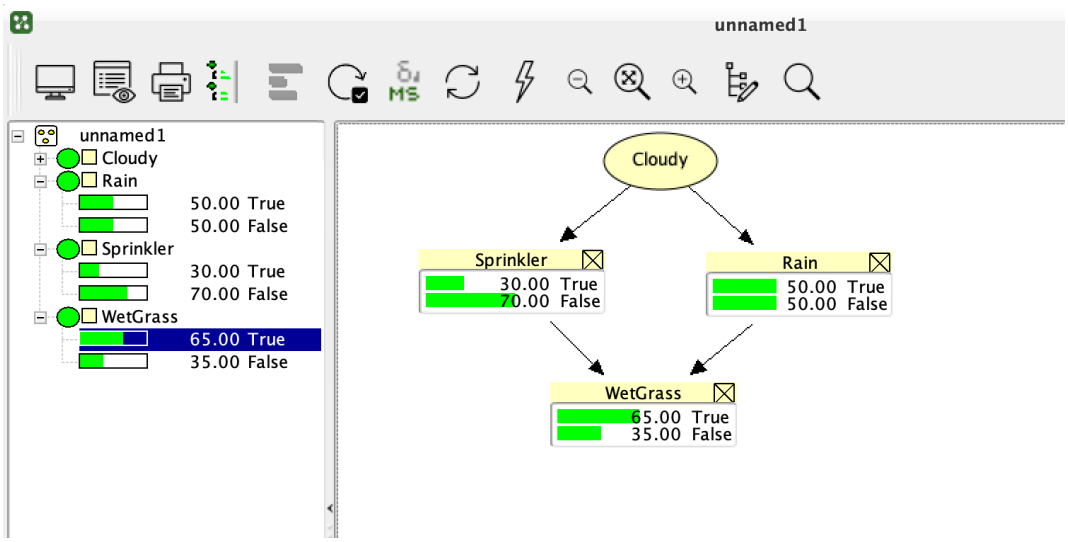

Figure 1: Bayes network implementation in Hugin-Lite showing prior marginals before any evidence is entered.

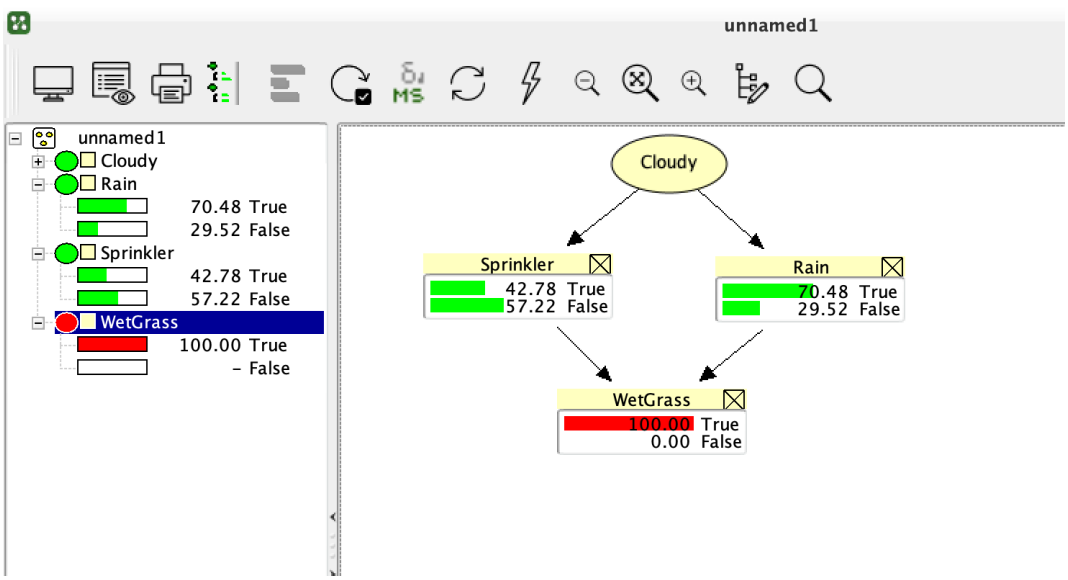

Figure 2: Posterior marginals after entering evidence $W{=}T$ (grass is wet). Rain becomes more likely than Sprinkler, matching analytical results.

Q11. Conditional Independencies and D-Separation in the Network

The network represents the joint distribution

$$P(C, S, R, W) = P(C)\, P(S \mid C)\, P(R \mid C)\, P(W \mid S, R),$$

which encodes the following conditional independencies through d-separation:

(i) S and R are conditionally independent given C. Both *Sprinkler* and *Rain* share the parent node *Cloudy* and have no direct edge between them:

$$S \perp\!\!\!\perp R \mid C.$$

Once we know whether it is cloudy, information about the sprinkler gives no additional information about rain.

(ii) C and W are independent given $\{S, R\}$. All influence of *Cloudy* on *WetGrass* is mediated through *Sprinkler* and *Rain*. Conditioning on both of these variables blocks all active paths:

$$C \perp\!\!\!\perp W \mid \{S, R\}.$$

(iii) Explaining away at collider W. Although S and R are independent given C, they become dependent when conditioned on their common effect W:

$$S \not\perp\!\!\!\perp R \mid W.$$

Observing that the grass is wet increases the probability of both possible causes (sprinkler or rain). If one cause is later confirmed, the probability of the other decreases—illustrating the "explaining away" phenomenon.

(iv) Summary. The graph combines three canonical causal patterns: (1) a *common cause* (C influencing S and R), (2) a *causal chain* ($C \to R \to W$), and (3) a *collider* at W receiving arrows from both S and R. These structural independencies justify the factorization used in Q8 and explain the posterior changes seen in Q9 and Q10.